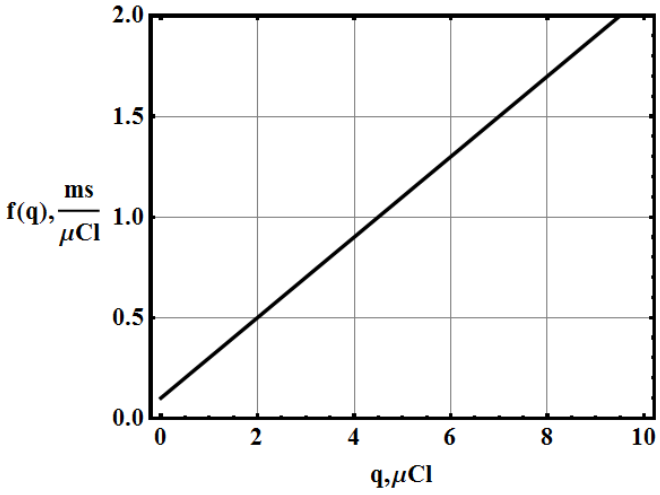


$$f(q) = a + bq, \quad (8)$$

$$a = 0.10\ \frac{\text{ms}}{\text{μCl}}, \quad (9)$$

$$b = 0.20\ \frac{ms}{(\text{μCl})^2}. \quad (10)$$

$q = 4$ мкКл (7)

(8)

$$t = aq + \frac{1}{2}bq^2 = 2.0\ ms \quad (11)$$

4. **[0.5점]** Δt $q_0 = 4$ μCl

$q = 8$ μCl ,

$$t = (q - q_0)\left(a + \frac{1}{2}b(q + q_0)\right) = 5.2ms. \quad (12)$$

5. **[0.5점]** (11)

$$q_{1/2} = \frac{-a \pm \sqrt{a^2 + 2bt}}{b}. \quad (13)$$

$q(0) = 0$,

(13) .

$$q = \frac{\sqrt{a^2 + 2bt} - a}{b} = 5{,}0\ \text{μCl}. \quad (14)$$

6. **[0.5점]** . ,

$$q = CU, \quad (15)$$

$$I = \frac{dq}{dt} = C\frac{dU}{dt} \sim \frac{dU}{dt}. \quad (16)$$

.

.

$U \sim \sin \omega t$ $I \sim \cos \omega t = \sin\left(\omega t - \frac{\pi}{2}\right)$,

,

$$\varphi = -\frac{\pi}{2}.$$

(16)

.

. ,

$$\varphi = -\frac{\pi}{2}. \quad (17)$$

7. **[4.0점]** .

. ,

$$U_C = 5{,}000\ V, \quad (18)$$

. ,